

and short-range UWB radar into one chip with an integrated MCU. This feature allows manufacturers to transform a UWB-based car access system into



a platform that supports multiple applications with the same hardware, reducing the need for additional systems and saving costs, space, and weight. The chip includes improved CPU performance, increased memory, and an integrated digital CAN transceiver to lower system costs.

NXP Semiconductors
<https://www.nxp.com>

Advanced system-on-chip for 5G small cell radio units

Picocom introduces the PC805, a system-on-chip (SoC) designed for 5G small cell Open RAN radio units



(O-RUs). Simplifying the development of 5G NR/LTE small cell O-RUs for enterprise, industrial, neutral host, and private

networks, the PC805 reduces the bill of materials compared to current designs. Connecting with O-DUs through Open Fronthaul (split 7.2) and allowing connections to RFICs via a JESD204B high-speed serial interface, this SoC can aggregate four or more 4T4R carriers in a 200MHz instantaneous bandwidth (IBW). It supports multiple bands for TDD and FDD in 5G NR and LTE, and using two SoCs in cascade can double the bandwidth.

Picocom
<https://picocom.com>

AI module transforming edge devices in multiple industries

DEEPM's DX-M1 module introduces intelligence to edge devices like robots in unmanned industrial sites, social infrastructure, and everyday life. The

company's AI chip source solutions offer advanced AI algorithm support for edge AI applications, providing AI accuracy comparable to GPUs with the highest effective power-to-performance

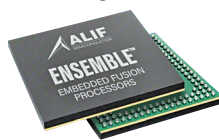


ratio globally. This technology significantly reduces the need for on-chip SRAM and off-chip DRAM, thereby cutting AI implementation costs. Target applications include consumer and industrial robots, machine vision systems, AI-integrated industrial PCs, high-performance computing, smart factories, edge computing solutions, and smart mobility initiatives. The AI Booster stands out as the world's most cost-efficient, power-efficient, and performance-efficient solution.

DEEPM
<https://deepm.ai>

AI-enhanced 32-bit MCUs and fusion processors

Alif Semiconductor announces the Ensemble product line, featuring 32-bit microcontrollers and fusion processors designed to outperform traditional



32-bit MCUs in AI model executions, significantly improving performance. For

example, incorporating a microNPU with a CPU core for an object detection model reduces inference time by 74 times compared to just the CPU core. Alif's CPU core increases AI/ML performance ten times more than standard MCU cores, leading to a total performance gain of up to 740 times. This performance increase does not compromise energy efficiency. The devices also boast security features, including read-out prevention to protect intellectual property, code originality and integrity checks to prevent malware attacks, and extensive cryptography to safeguard data on the device and in communication channels.

Alif Semiconductor
<https://alifsemi.com>

Industry's first 800G DSP IC to boost connectivity

Credo Technology Group Holding Ltd unveils the Dove 850 800G digital signal processor (DSP) IC, a groundbreaking industry-first. With the rising prevalence of AI deployments in hyperscale data centres, this DSP plays a pivotal



role in connecting GPUs, enhancing system performance, and reducing costs. It empowers transceiver

vendors to choose from various optics based on application requirements, such as VCSELs and silicon photonics. The DSP improves optical transceivers by ensuring compliance, minimising manual tuning, and enhancing performance in different conditions and with various components. It features independent phase-locked loops (PLLs) for each lane, making it suitable for breakout applications.

Credo Technology Group Holding Ltd
<https://credosemi.com>

Microcontroller boosts wireless device efficiency

STMicroelectronics launches a new microcontroller (MCU) that merges wireless device design expertise with the STM32 architecture. The STM-



32WL3 wireless MCUs, designed for remote applications such as metering,

monitoring devices, and data collection in buildings, factories, and cities, aim to lower power consumption and streamline activities. These MCUs enable rapid product development, featuring a multi-protocol radio for efficient, long-range communication in ISM frequency bands. Equipped with an LCD controller and LC sensor controller for fluid flow measurement, the MCU extends battery life, contributing to an improved user experience, service, and reduced environmental impact.

STMicroelectronics
<https://www.st.com>